

David Joseph N. Cortez

Brgy. Puting Kahoy Silang, Cavite | 09662565417 | djnc61003@gmail.com



<https://www.linkedin.com/in/david-joseph-cortez/>



<https://github.com/djncortez>



<https://djncortez.netlify.app/>

SKILLS

Languages: Python, Java, Kotlin

Frontend: HTML, CSS, JavaScript

Data and Analytics: Pandas, NumPy, Matplotlib

Database: PostgreSQL, MySQL

Tools: VS Code, Git, Android Studio, Supabase, Power BI

AI Tools: Claude, Gemini, ChatGPT, Copilot, VO

PROFESSIONAL EXPERIENCE

Ready, Set, Kinder

Educational Mobile App

- Developed a multi-role Android app (Admin, Teacher, Parent, Student) for monitoring early childhood education for special needs learners, featuring class management, bulk CSV import, and role-based access control.
- Built a Teacher Dashboard with per-student game performance metrics and visual score breakdowns across Literacy, Cognitive, and Numeracy categories, scoped to each teacher's enrolled classes.
- Integrated Supabase (PostgreSQL, Auth, RLS) with a dual-client architecture for secure admin operations, and Groq AI (LLaMA 3.3-70B) for automated student performance report generation.

EDUCATION

Mapua Malayan Colleges Laguna | 2022-2026

Bachelor of Science in Computer Science

- Specialization in Data Science and Analytics
- GPA: 1.57

PROJECTS

Experimental Visual Playground (VOID.LAB) – HTML / CSS / JavaScript

- Built an interactive generative art portfolio featuring three real-time canvas experiments: a magnetic particle repulsion field, a Perlin noise flow field, and a physics-based type disintegration system.
- Implemented all physics simulations, vector field algorithms, and noise functions from scratch using the HTML5 Canvas API and vanilla JavaScript — no external libraries or frameworks.

Flight Price Trend Analyzer – Python / Pandas / NumPy / Matplotlib

- Built a dataset of 13,000+ synthetic flight price records for Philippine routes, simulating real-world pricing factors like holidays, booking windows, and day-of-week patterns.
- Analyzed pricing trends using Pandas to surface insights such as the cheapest travel days and optimal booking timing (90–120 days in advance).
- Created 4 data visualizations including heatmaps, trend lines, and violin plots, plus an interactive HTML dashboard to explore results in the browser.